

Software design & developement

21/05/2025

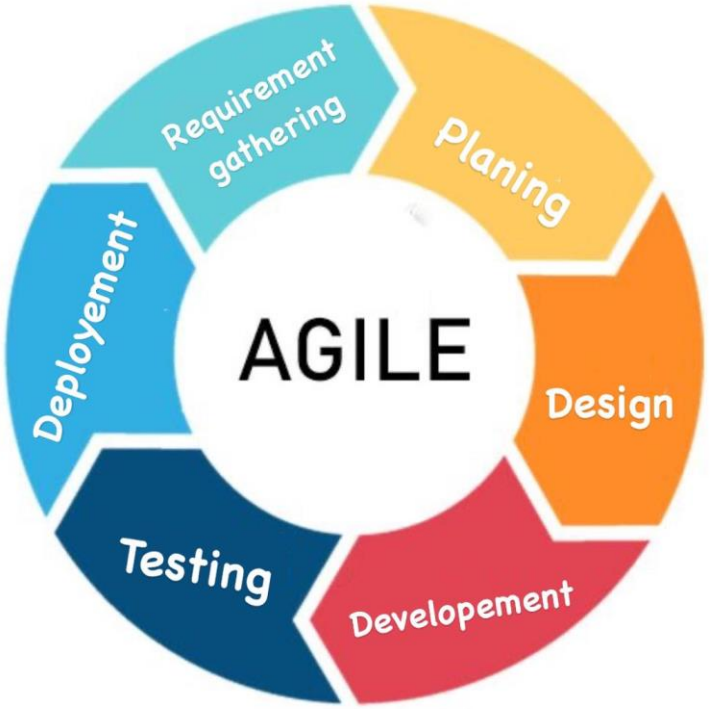
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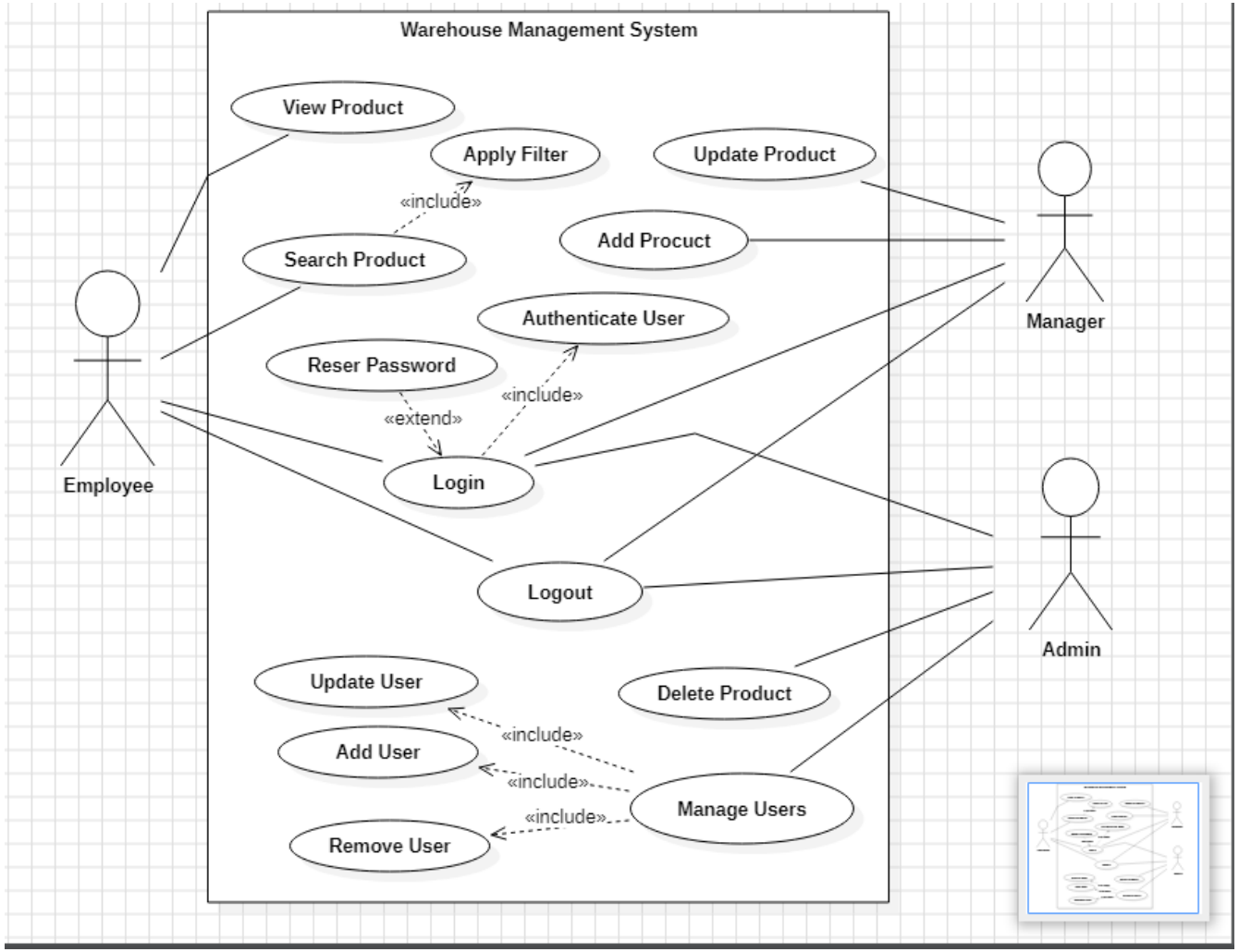
YEAR: 2024/2025

INSTITUTE: SIST

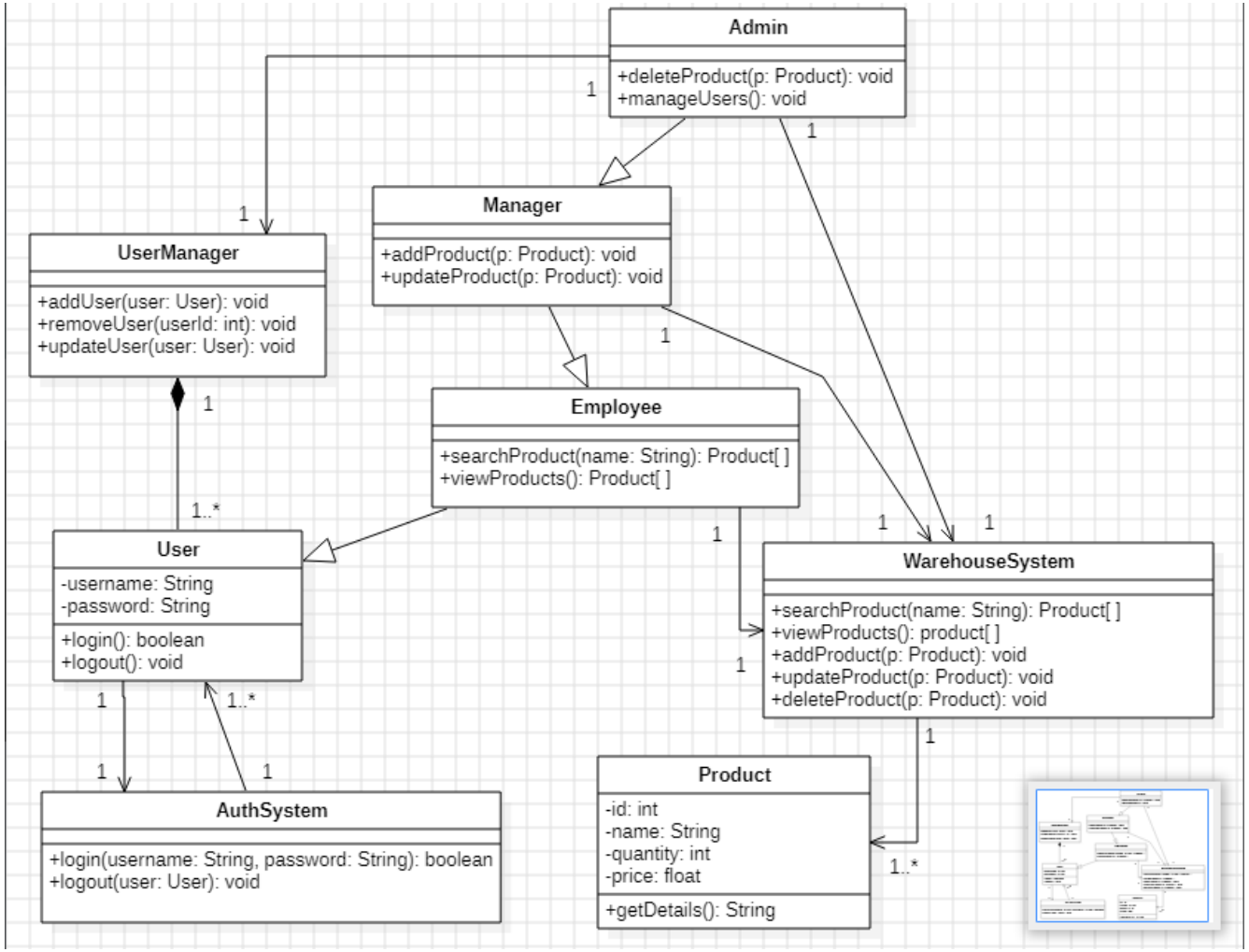
Sdlc agile lifesycle



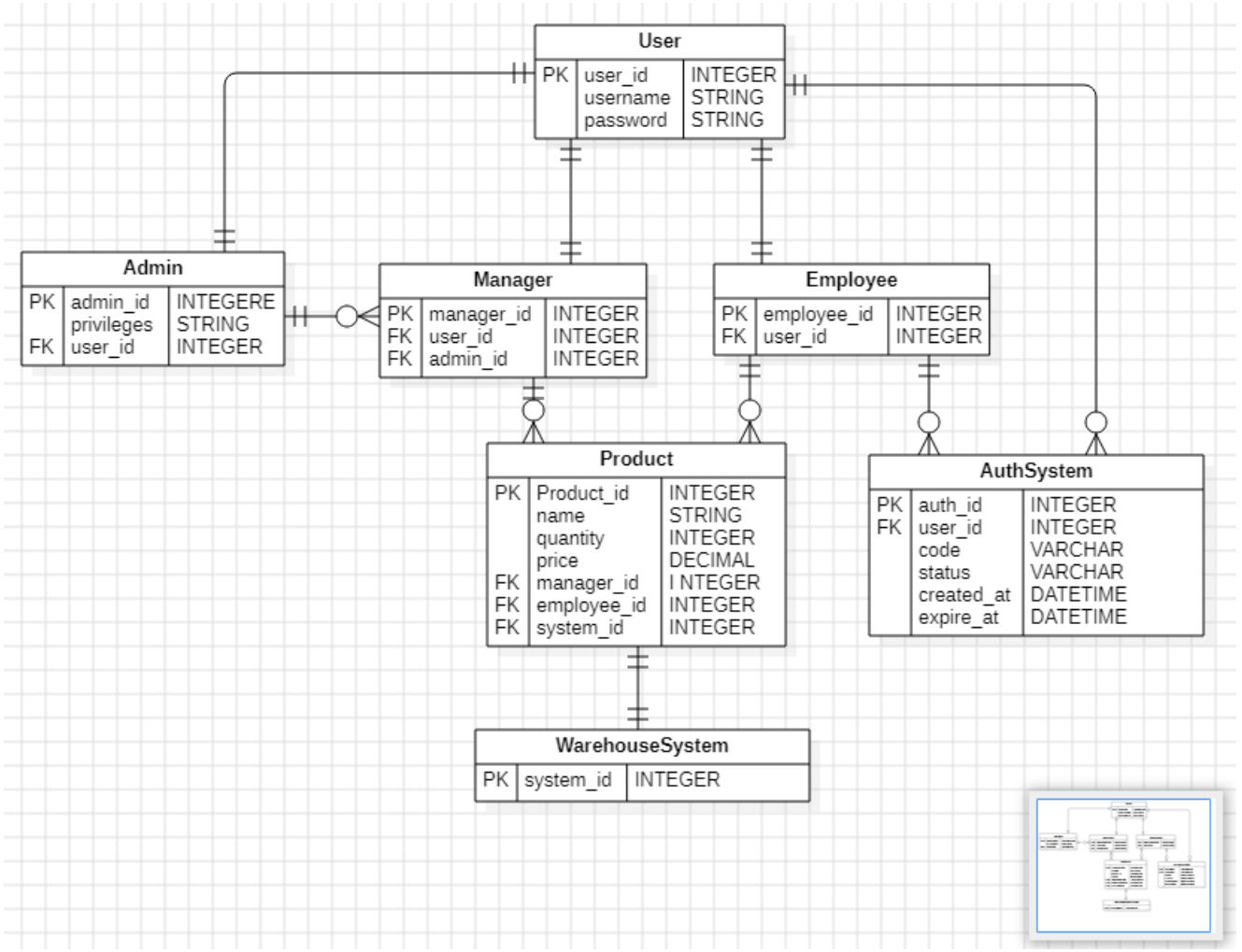
Use case diagram

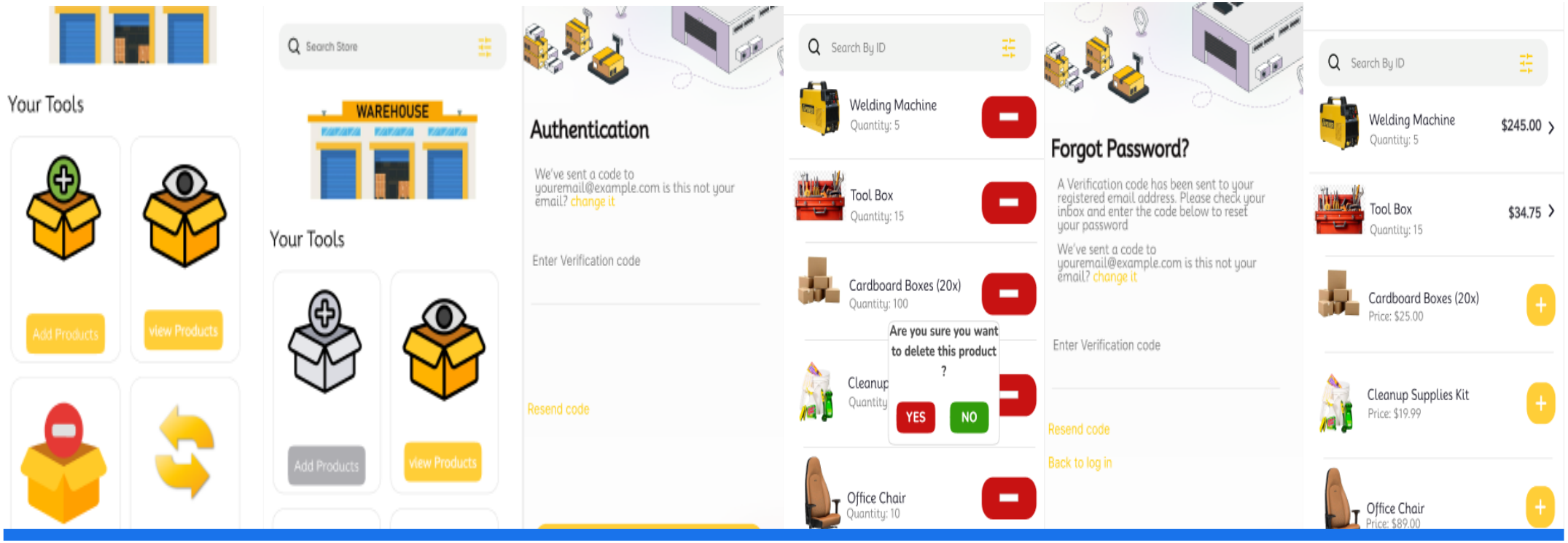


Uml class diagram

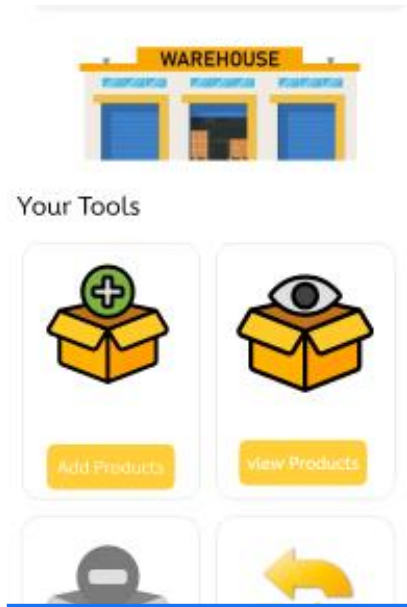


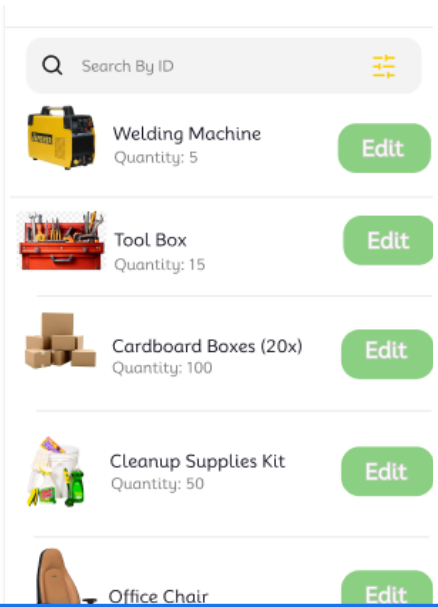
Realationship entities



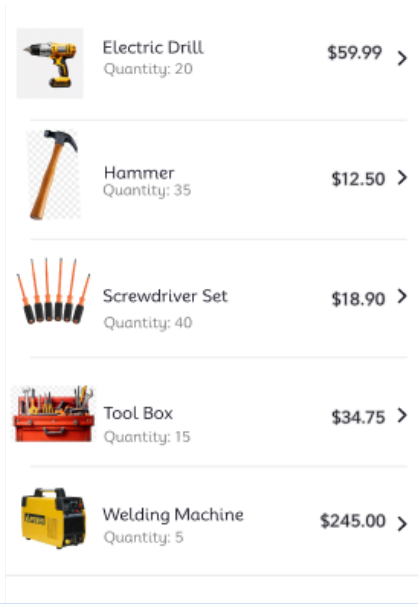


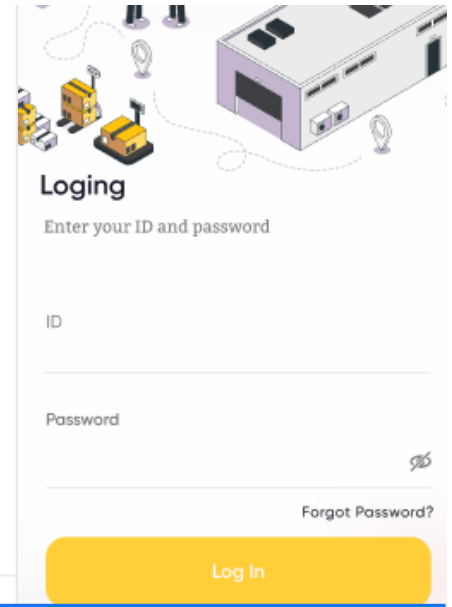
Front end design


WAREHOUSE
Your Tools
Add Products
view Products


Search By ID
Welding Machine
Quantity: 5
Edit
Tool Box
Quantity: 15
Edit
Cardboard Boxes (20x)
Quantity: 100
Edit
Cleanup Supplies Kit
Quantity: 50
Edit
Office Chair
Edit


Search By ID
User ID
Username
Role
101 amira Employee
102 karim Manager
103 yassir Admin
104 laila Employee
105 soufiane Manager
106 hanaa Employee
107 nabil Employee
108 asmae Manager
109 reda Admin
110 fatima Employee
111 mehdi Manager
112 salma Employee


Electric Drill
Quantity: 20
$59.99
Hammer
Quantity: 35
$12.50
Screwdriver Set
Quantity: 40
$18.90
Tool Box
Quantity: 15
$34.75
Welding Machine
Quantity: 5
$245.00


Loging
Enter your ID and password
ID
Password
Forgot Password?
Log In

9:41

Manage Users Section

Search By ID

User ID	Username	Role	
101	amira	Employee	✅
102	karim	Manager	
103	yassir	Admin	
104	laila	Employee	
105	soufiane	Manager	
106	hanaa	Employee	✅
107	nabil	Employee	✅
108	asmae	Manager	
109	reda	Admin	
110	fatima	Employee	
111	mehdi	Manager	
112	salma	Employee	✅

Remove

Update

Add



Sign Up

Enter your credentials to continue

ID

Email

Password

Confirm Password

By continuing you agree to our Terms of Service and Privacy Policy.

Sing Up

Already have an account? Singup

Databases

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	manager_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	User_Id	int			No	None			Change Drop More
3	admin_id	int			No	None			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	system_id	int			No	None		AUTO_INCREMENT	Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	User_Id	int			No	None		AUTO_INCREMENT	Change Drop More
2	Username	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
3	Password	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	employee_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	User_Id	int			No	None			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	admin_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	privilages	varchar(255)	utf8mb4_0900_ai_ci		No	None			Change Drop More
3	User_Id	int			No	None			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	product_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	name	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
3	quantity	int			No	None			Change Drop More
4	price	int			No	None			Change Drop More
5	manager_id	int			No	None			Change Drop More
6	employee_id	int			No	None			Change Drop More
7	system_id	int			No	None			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	auth_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	User_Id	int			No	None			Change Drop More
3	code	varchar(20)	utf8mb4_0900_ai_ci		No	None			Change Drop More
4	status	varchar(20)	utf8mb4_0900_ai_ci		No	None			Change Drop More
5	created_at	datetime			No	None			Change Drop More
6	expires_at	datetime			No	None			Change Drop More

```php
<?php
$host = 'localhost';
$db   = 'warehouse_management';
$user = 'root';
$pass = 'Reem1234@';
$charset = 'utf8mb4';

$dsn = "mysql:host=$host;dbname=$db;charset=$charset";
$options = [
    PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
];

try {
    $pdo = new PDO($dsn, $user, $pass, $options);
} catch (\PDOException $e) {
    die('Database connection failed: ' . $e->getMessage());
}
?>
```

Database link

```
} else {
    echo "No account found with that email.";
}
```

```
} else {
    echo "Invalid ID or password.";
}
```

```
$hashed_password = password_hash($password, PASSWORD_DEFAULT);


$stmt = $pdo->prepare("INSERT INTO user (user_id, email, password) VALUES (?, ?, ?)");
$stmt->execute([$user_id, $email, $hashed_password]);


$stmt = $pdo->prepare("INSERT INTO $role (user_id) VALUES (?)");
$stmt->execute([$user_id]);

echo "Registration successful!";
```

```
<?php
session_start();
require_once('../config/db.php');

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $user_id = $_POST['user_id'] ?? '';
    $password = $_POST['password'] ?? '';

    if (empty($user_id) || empty($password)) {
        die('Please enter both User ID and Password.');
    }


    $stmt = $pdo->prepare("SELECT * FROM user WHERE user_id = ?");
    $stmt->execute([$user_id]);
    $user = $stmt->fetch();

    if ($user && password_verify($password, $user['password'])) {
        $_SESSION['user_id'] = $user['user_id'];


        $role = 'Unknown';

        $stmt = $pdo->prepare("SELECT * FROM admin WHERE user_id = ?");
        $stmt->execute([$user_id]);
        if ($stmt->fetch()) $role = 'admin';

        $stmt = $pdo->prepare("SELECT * FROM manager WHERE user_id = ?");
        $stmt->execute([$user_id]);
        if ($stmt->fetch()) $role = 'manager';

        $stmt = $pdo->prepare("SELECT * FROM employee WHERE user_id = ?");
```

```
    $stmt = $pdo->prepare("SELECT * FROM employee WHERE user_id = ?");
    $stmt->execute([$user_id]);
    if ($stmt->fetch()) $role = 'employee';

    $_SESSION['role'] = $role;


    $generated_code = rand(100000, 999999);
    $_SESSION['verification_code'] = $generated_code;
    $_SESSION['verify_action'] = 'login';

    // Send the code to the email
    mail($user['email'], "Login Verification Code", "Your verification code is: $generated_code");


    exit;
} else {
    echo "Invalid ID or password.";
}
}
?>
```

Backend , loging,signup

```php
<?php
session_start();

$_SESSION = [];

if (ini_get("session.use_cookies")) {
    $params = session_get_cookie_params();
    setcookie(session_name(), '', time() -
        $params["path"], $params["domain"],
        $params["secure"], $params["httpon
    );
}

session_destroy();

header("Location:loging.php");
exit;
?>
```

```php
<?php
require_once('../config/db.php');

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $user_id = $_POST['user_id'] ?? '';
    $email = $_POST['email'] ?? '';
    $password = $_POST['password'] ?? '';
    $confirm_password = $_POST['confirm_password'] ?? '';
    $role = $_POST['role'] ?? '';

    // can't keep the fields empty
    if (empty($user_id) || empty($email) || empty($password) || empty($confirm_password) || empty($role)) {
        die("All fields are required.");
    }

    // password match confirm password
    if ($password !== $confirm_password) {
        die("Passwords do not match.");
    }

    // role selection
    $allowed_roles = ['admin', 'manager', 'employee'];
    if (!in_array($role, $allowed_roles)) {
        die("Invalid role selected.");
    }

    // check if the user already exist
    $stmt = $pdo->prepare("SELECT * FROM user WHERE user_id = ? OR email = ?");
    $stmt->execute([$user_id, $email]);
    if ($stmt->fetch()) {
        die("User ID or email already exists.");
    }
```

```php
    $stmt = $pdo->prepare("SELECT * FROM manager WHERE user_id = ?");
    $stmt->execute([$user_id]);
    if ($stmt->fetch()) {
        $role = 'manager';
    }


    $stmt = $pdo->prepare("SELECT * FROM employee WHERE user_id = ?");
    $stmt->execute([$user_id]);
    if ($stmt->fetch()) {
        $role = 'employee';
    }

    $_SESSION['role'] = $role;

    // lead to the page based on the id and the role
    switch ($role) {
        case 'admin':
            header("Location: ../admin_dashboard.php");
            break;
        case 'manager':
            header("Location: ../manager_dashboard.php");
            break;
        case 'employee':
            header("Location: ../employee_dashboard.php");
            break;
        default:
            echo "Role not recognized.";
            break;
    }
```

```php
    $email = $_POST['email'] ?? '';

    if (empty($email)) {
        die("Please enter your email.");
    }


    $stmt = $pdo->prepare("SELECT * FROM user WHERE email = ?");
    $stmt->execute([$email]);
    $user = $stmt->fetch();

    if ($user) {
        $_SESSION['user_id'] = $user['user_id'];

        // this part generate a verification code and store it
        $generated_code = rand(100000, 999999);
        $_SESSION['verification_code'] = $generated_code;
        $_SESSION['verify_action'] = 'reset_password';
```



```php
<?php
session_start();
require_once('../config/db.php');


if (!isset($_SESSION['user_id'])) {
    die("Access denied.");
}


$stmt = $pdo->query("SELECT * FROM product");
$products = $stmt->fetchAll();

?>
```

```php
<?php
session_start();
require_once('../config/db.php');

if (!isset($_SESSION['user_id']) || !in_array($_SESSION['role'], ['admin', 'manager'])) {
    die("Access denied.");
}

$stmt = $pdo->query("SELECT * FROM product");
$existing_products = $stmt->fetchAll();

?>
```

```php
<?php
session_start();
require_once('../config/db.php');


if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $product_id = $_POST['product_id'] ?? '';

    if (empty($product_id)) {
        die("No product selected.");
    }

    // Delete the product
    $stmt = $pdo->prepare("DELETE FROM product WHERE product_id = ?");
    $stmt->execute([$product_id]);


    header("Location: list_products_for_delete.php");
    exit;
}
?>
```

```php
<?php
session_start();
require_once('../config/db.php');


if (!isset($_SESSION['user_id']) || !in_array($_SESSION['role'], ['admin', 'manager'])) {
    die("Access denied.");
}


$product_id = $_GET['id'] ?? null;
if (!$product_id) {
    die("Invalid request.");
}


$stmt = $pdo->prepare("SELECT * FROM product WHERE id = ?");
$stmt->execute([$product_id]);
$product = $stmt->fetch();

if (!$product) {
    die("Product not found.");
}

?>
```

```php
<?php
session_start();
require_once('../config/db.php');


if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}


$stmt = $pdo->query("SELECT * FROM product");
$products = $stmt->fetchAll();
?>
```

Products, view,add,update,delete



```
_start();
e_once('../config/db.php');


sset($_SESSION['user_id']) || !in_array($_SESSION['role'], ['admin'
e("Access denied.");


SERVER['REQUEST_METHOD'] === 'POST') {
roduct_id = $_POST['id'] ?? null;
ame = $_POST['name'] ?? '';
uantity = $_POST['quantity'] ?? '';
rice = $_POST['price'] ?? '';

 (!$product_id || empty($name) || empty($quantity) || empty($price)
  die("All fields are required.");



tmt = $pdo->prepare("UPDATE product SET name = ?, quantity = ?, pri
tmt->execute([$name, $quantity, $price, $product_id]);


ader("Location: update_product.php");
it;
```

```php
<?php
session_start();
require_once('../config/db.php');


if (!isset($_SESSION['user_id']) || !in_array($_SESSION['role'], ['admin',
    die("Access denied.");
}


$stmt = $pdo->query("SELECT * FROM product");
$products = $stmt->fetchAll();


?>
```

```php
ion_start();
ire_once('../config/db.php');

nly admin and manager who are responsible
!isset($_SESSION['user_id']) || !in_array($_SESSION['role'], ['ad
die("Access denied.");


$_SERVER['REQUEST_METHOD'] === 'POST') {
$name = $_POST['name'] ?? '';
$quantity = $_POST['quantity'] ?? '';
$price = $_POST['price'] ?? '';

if (empty($name) || empty($quantity) || empty($price)) {
    die("All fields are required.");
}

// add new product
$stmt = $pdo->prepare("INSERT INTO product (name, quantity, price
$stmt->execute([$name, $quantity, $price]);


header("Location: add_product.php");
exit;
```

Users management,add,delete,update

```php
<?php
session_start();
require_once('../config/db.php');


if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}

$stmt = $pdo->query("SELECT * FROM user");
$users = $stmt->fetchAll();


?>
```

```php
<?php
session_start();
require_once('../config/db.php');

if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $user_ids = $_POST['user_ids'] ?? [];

    if (empty($user_ids) || !is_array($user_ids)) {
        die("No users selected.");
    }

    foreach ($user_ids as $user_id) {
        // Delete from all role tables
        $pdo->prepare("DELETE FROM admin WHERE user_id = ?")->execute([$user_id]);
        $pdo->prepare("DELETE FROM manager WHERE user_id = ?")->execute([$user_id]);
        $pdo->prepare("DELETE FROM employee WHERE user_id = ?")->execute([$user_id]);

        // Delete from user table
        $pdo->prepare("DELETE FROM user WHERE user_id = ?")->execute([$user_id]);
    }

    header("Location: manage_users.php");
    exit;
}
?>
```

```php
<?php
session_start();
require_once('../config/db.php');

if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $user_id = $_POST['user_id'] ?? '';
    $email = $_POST['email'] ?? '';
    $password = $_POST['password'] ?? '';
    $role = $_POST['role'] ?? 'employee'; // default role can be changed by selection

    if (empty($user_id) || empty($email) || empty($password)) {
        die("All fields are required.");
    }


    $hashed_password = password_hash($password, PASSWORD_DEFAULT);


    $stmt = $pdo->prepare("INSERT INTO user (user_id, email, password) VALUES (?, ?, ?)");
    $stmt->execute([$user_id, $email, $hashed_password]);


    if ($role === 'admin') {
        $stmt = $pdo->prepare("INSERT INTO admin (user_id) VALUES (?)");
        $stmt->execute([$user_id]);
    } elseif ($role === 'manager') {
        $stmt = $pdo->prepare("INSERT INTO manager (user_id) VALUES (?)");
        $stmt->execute([$user_id]);
```

```php
<?php
session_start();
require_once('../config/db.php');

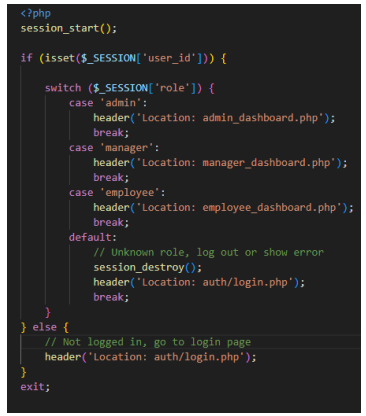
if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $user_id = $_POST['user_id'] ?? '';
    $email = $_POST['email'] ?? '';
    $password = $_POST['password'] ?? '';
    $role = $_POST['role'] ?? '';

    if (empty($user_id) || empty($email) || empty($role)) {
        die("User ID, email, and role are required.");
    }

    if (!empty($password)) {
        $hashed_password = password_hash($password, PASSWORD_DEFAULT);
        $stmt = $pdo->prepare("UPDATE user SET email = ?, password = ? WHERE user_id = ?");
        $stmt->execute([$email, $hashed_password, $user_id]);
    } else {
        $stmt = $pdo->prepare("UPDATE user SET email = ? WHERE user_id = ?");
        $stmt->execute([$email, $user_id]);
    }

    // Update role, delete old and add new
    $pdo->prepare("DELETE FROM admin WHERE user_id = ?")->execute([$user_id]);
    $pdo->prepare("DELETE FROM manager WHERE user_id = ?")->execute([$user_id]);
    $pdo->prepare("DELETE FROM employee WHERE user_id = ?")->execute([$user_id]);
```



```php
<?php
session_start();

if (isset($_SESSION['user_id'])) {

    switch ($_SESSION['role']) {
        case 'admin':
            header('Location: admin_dashboard.php');
            break;
        case 'manager':
            header('Location: manager_dashboard.php');
            break;
        case 'employee':
            header('Location: employee_dashboard.php');
            break;
        default:
            // Unknown role, log out or show error
            session_destroy();
            header('Location: auth/login.php');
            break;
    }
} else {
    // Not logged in, go to login page
    header('Location: auth/login.php');
}
exit;
```



```php
<?php
session_start();
require_once('config/db.php');


if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    die("Access denied.");
}


$stmt = $pdo->query("SELECT COUNT(*) FROM user");
$total_users = $stmt->fetchColumn();


$stmt = $pdo->query("SELECT COUNT(*) FROM product");
$total_products = $stmt->fetchColumn();


?>
```



```php
<?php
session_start();
require_once('config/db.php');

if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'employee') {
    die("Access denied.");
}


?>
```

Users dashboard, admin, manager, employee

```php
<?php
session_start();

if (isset($_SESSION['user_id'])) {

    switch ($_SESSION['role']) {
        case 'admin':
            header('Location: admin_dashboard.php');
            break;
        case 'manager':
            header('Location: manager_dashboard.php');
            break;
        case 'employee':
            header('Location: employee_dashboard.php');
            break;
        default:
            // Unknown role, log out or show error
            session_destroy();
            header('Location: auth/login.php');
            break;
    }
} else {
    // Not logged in, go to login page
    header('Location: auth/login.php');
}
exit;
```

Index page

References

Agile Alliance (n.d.) *Agile Glossary*. Available at: https://www.agilealliance.org (Accessed: 16 May 2025).

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